Customer Personality Analysis Dashboard

CUSTOMER PERSONALITY ANALYTICS

An Interactive Business Intelligence Dashboard

Built with Microsoft Power BI Desktop

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- Tool Used Microsoft Power BI Desktop
- Dataset Customer Personality Analysis
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1. Problem Statement

Customer-focused businesses collect large amounts of data on demographics, income, household structure, complaints, product preferences, and purchasing channels. But rows and columns alone don't explain what matters. The challenge is converting raw customer data into answers that support targeting, category strategy, and segment prioritization — quickly enough to act.

This dashboard was built to summarize customer demographics, purchasing channels, product category mix, complaint rates, and family composition so stakeholders can understand who the customers are, how they buy, and what they prefer in a single screen.

2. Business Use (Real-World Purpose)

This dashboard supports practical decision-making by helping teams identify high-value customer segments, understand purchasing behavior across

channels, and align product and offer strategy with income and demographic patterns.

2. Solution & Dashboard Features

The finished output is a single-page interactive Power BI report. Each visual supports fast comparison across segments, and the dashboard is designed to be readable in under two minutes.

Below is a breakdown of the key components and what each one reveals.

Average Income by Education — Bar Chart

Compares average income across education levels:

- PhD: 56K
- Master: 53K
- Graduation: 53K
- 2n Cycle: 48K
- Basic: 20K

This enables income-based targeting and highlights a sharp separation between the Basic group and higher-education groups.

Average Income by Marital Status — Bar Chart

Compares average income across marital-status groups:

- Absurd: 72K
- Widow: 56K
- Together: 53K
- Divorced: 53K
- Married: 52K
- Single: 51K

- YOLO: 48K

The “Absurd” label is anomalous and should be validated for data quality.

Customers by Year of Birth — Distribution

Shows the customer age profile (year-of-birth distribution) to understand generational composition.

Customers by Marital Status and Kids at Home — Column Chart

Shows family composition by marital segment:

- Married: 394
- Together: 261
- Single: 223
- Divorced: 96
- Widow: 18
- Alone: 3
- Absurd: 0
- YOLO: 0

This highlights that Married drives the largest share of households with kids, with Together and Single representing major additional reach.

Customers by Education — Pie Chart

Education mix:

- Graduation: 50.31%
- PhD: 21.70%
- Master: 16.52%
- 2n Cycle: 9.06%
- Basic: ~2–3%

This indicates messaging should generally favor higher-education audiences.

Customers by Complain — Donut Chart

Complaint rate:

- No: 99.06%
- Yes: 0.94%

Customer by Marital Status — Donut Chart

Marital-status composition:

- Married: 38.57%
- Together: 25.89%
- Single: 21.43%
- Divorced: 10.36%
- Widow: 3.44%

Purchasing Platforms — Pie Chart

Channel mix:

- Store purchases: 46.18%
- Web purchases: 32.58%
- Catalog purchases: 21.23%

Products — Pie Chart

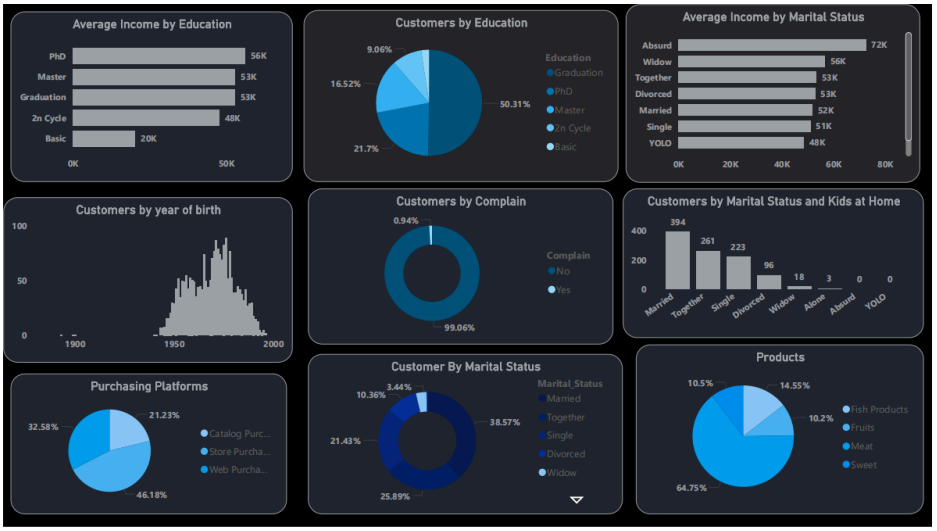
Product-category contribution:

- Meat: 64.75%
- Fish Products: 14.55%
- Sweet: 10.50%
- Fruits: 10.20%

This reveals heavy concentration in Meat and a risk if Meat demand or supply shifts; promoting underrepresented categories could balance contribution.

3. Dashboard Screenshot

Figure 1 — Customer Personality Analysis Dashboard Overview, Power BI Desktop



4. Tech Stack & Dataset

Area	Details
Visualization Tool	Microsoft Power BI Desktop (2024)
Data Import	Dataset already prepared; loaded into Power BI
Transformation	Power Query Editor (M Language)
Calculations	DAX measures (averages, counts, proportions)
Chart Types	Bar, Donut, Pie, Distribution
Interactivity	Cross-filtering across visuals
Project File	.pbix (native Power BI format)

5. Key Insights & What Stands Out

a) The Married segment drives the largest share of households with kids (394), with Together (261) and Single (223) following, while tiny groups like Alone (3), Widow (18) and Absurd/YOLO (0) are negligible.

b) Targeting the Together and Single segments could significantly expand reach beyond the Married-heavy base.

c) Meat dominates product contribution (65%), far surpassing Fish Products (15%), Sweet (11%), and Fruits (10%), indicating heavy category concentration.

d) There is a risk to revenue if Meat demand or supply shifts; promoting underrepresented categories could balance contribution.

e)Education profile is skewed: Graduation ~50%, PhD ~22%, Master ~16%, with Basic only ~2–3%, suggesting messaging should favor higher-education audiences.

f) PhD holders have the highest average income (56K), followed by Graduation (53K) and Master (53K); Basic is substantially lower (20K), enabling income-based targeting.

g) The large gap between Basic and higher education groups implies education-based segmentation can support premium product targeting.

h)Anomalous label “Absurd” in Income by Marital Status shows very high income (72K) and warrants validation for data quality.

i) Among valid marital-status groups, Widow (56K) and Divorced/Together (53K) have higher or comparable incomes to Married (52K) and Single (51K), indicating non-married groups may offer similar or greater purchasing power.

6. Future Scope & Conclusion

Where This Dashboard Could Go Next

i)Add drill-through pages by marital status and education to surface product/channel preference by segment.
ii) Add KPI cards (Total customers, Average income, Complaint rate, Kids at home) for faster top-level scanning.
iii) Validate categorical anomalies (e.g., Absurd, YOLO) and document data-cleaning assumptions.
iv) Extend the model with campaign response or purchase history (if available) for lifecycle analysis.

Closing Thoughts

Starting from customer-level demographic and purchase attributes and arriving at a single-screen dashboard was the central goal of this project. The family-

composition concentration (Married, Together, Single), the product concentration (Meat), and the clear income separation by education become obvious once visualized. That is the value of good BI: it makes what matters impossible to miss.